FINAL PROJECT REPORT

Global AI Adoption in Education

Data Analytics • Tableau Visualization • Flask Web Integration

Project	Global AI Adoption in Education
Repository	github.com/PrasadMhasal10/Global_AI_Adoption_In_Education
Tools	Python, Flask, PostgreSQL, DBeaver, Tableau, HTML, CSS, JavaScript, GitHub
Prepared by	[Add your name and student ID]
Institution / Faculty	[Add institution and faculty details]

Report prepared using the project information available at the time of writing. Confirm the final workbook contents, dataset fields, and live links before submission.

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Note: Where exact implementation counts or verified metrics were not supplied, this report describes the intended analysis and leaves values for confirmation rather than inventing results.

1. Introduction

1.1 Project Overview

Global AI Adoption in Education is a data analytics project designed to explore patterns in the adoption and use of artificial intelligence (AI) in educational settings. The project combines a structured dataset, data storage and inspection, interactive Tableau visualizations, a Tableau Story, and a Flask-based web interface. The goal is to make the dataset easier to explore and to present analytical views in a clear, accessible format.

1.2 Objectives

- Organize education-related AI adoption data in a structured dataset.
- Inspect and prepare the data for analysis using CSV and database tools.
- Build Tableau visualizations to compare adoption, usage, impact, infrastructure, and barriers.
- Use filters and calculated fields to support interactive exploration.
- Present the dashboard and story through a Flask website.
- Maintain project code and supporting files in a public GitHub repository.

Project scope

The current project is a demonstration of an end-to-end analytics workflow. The dataset has been described as sample or synthetic data; therefore, the visualizations are suitable for demonstrating analytical methods, but should not be interpreted as verified global statistics unless the dataset is replaced with validated sources.

2. Project Initialization and Planning Phase

2.1 Define Problem Statement

Educational AI data can contain several dimensions—such as geography, year, education level, user group, tools, usage, infrastructure, and adoption barriers. Without a structured presentation, it can be difficult to compare these dimensions or communicate patterns. The project addresses this by organizing the data and providing a dashboard, story, and web interface for exploration.

2.2 Project Proposal (Proposed Solution)

The proposed solution is to use a CSV dataset as the data input, inspect or manage records with PostgreSQL and DBeaver where applicable, create charts and an interactive dashboard in Tableau, build a Tableau Story to guide the viewer through selected questions, and use Flask to provide a single web page with navigation buttons for the dashboard and story.

2.3 Initial Project Planning

Phase	Planned work	Output
Data preparation	Review CSV fields and check data consistency.	Prepared dataset
Database work	Import or inspect records using PostgreSQL/DBeaver, if used in the final workflow.	Queryable data
Visualization	Build charts, calculated fields, and filters in Tableau.	Dashboard
Storytelling	Arrange selected visualizations into a guided sequence.	Tableau Story
Web integration	Use Flask and front-end files to display Dashboard and Story options.	Web interface
Publishing	Store source files and documentation in GitHub; publish a narrated demo video separately.	Repository and demo URL

3. Data Collection and Preprocessing Phase

3.1 Data Collection Plan and Raw Data Sources Identified

The primary input described for this project is a CSV dataset containing education and AI-adoption-related fields. The dataset is used for analysis and visualization. A packaged Tableau workbook (.twbx) is also part of the project materials. The final submission should identify whether the dataset is synthetic, publicly sourced, or collected from a named source, and provide a source citation if applicable.

3.2 Data Quality Report

A data quality review should check missing values, duplicate rows, field types, inconsistent category labels, invalid numeric values, and year formatting. Exact quality counts were not provided for this report and should be filled in after checking the final CSV.

Check	What to verify	Result
Missing values	Count blanks in important fields.	[Fill in after checking]
Duplicate rows	Check for repeated records.	[Fill in after checking]
Data types	Confirm years are numeric/date-like and measures are numeric.	[Fill in after checking]
Category consistency	Standardize spelling and category labels.	[Fill in after checking]

3.3 Data Exploration and Preprocessing

The fields described for the project include Country, Region, Year, Education Level, User Type, AI Tool, AI Adoption Rate, AI Usage Hours, Student Count, Educator Count, Infrastructure Score, AI Impact Score, and Barrier. These columns support geographic, temporal, group-based, and tool-based comparisons. Before analysis, confirm that numeric measures are stored as numeric values and that category fields use consistent labels.

If the dataset is synthetic or sample data, that limitation should be stated in the dashboard, report, and presentation. Do not describe sample values as real-world evidence.

4. Data Visualization

4.1 Framing Business Questions

- How does the AI adoption rate vary across the countries represented in the dataset?

- How does adoption change across the years included in the data?
- How do AI usage hours differ by education level and user type?
- How do impact scores compare across the AI tools represented?
- How does infrastructure score vary by region?
- Which barriers appear in the dataset, and how are they associated with adoption?

4.2 Developing Visualizations

Visualization	Purpose
AI Adoption Rate by Country	Compare adoption values across represented countries.
AI Adoption Rate Trend by Year	Inspect changes across the years present in the dataset.
Average AI Usage Hours by Education Level	Compare usage across education levels.
AI Impact Score by AI Tool	Compare impact scores for the tools in the dataset.
AI Adoption Rate by User Type	Compare user groups, such as students and educators.
Infrastructure Score by Region	Explore regional differences in infrastructure scores.
AI Adoption Rate by Barrier	Review how barriers are represented alongside adoption.
Student vs Educator Usage and Adoption	Compare student and educator measures.

5. Dashboard

5.1 Dashboard Design File

The Tableau Dashboard brings the main visualizations together on one screen so that a viewer can compare multiple aspects of AI adoption without opening each chart separately. The intended interactive filters include Year, Country, Region, and User Type. Users can select filter values to focus the analysis on a subset of records.

The project description also identifies calculated fields named Adoption Category and Impact Category, intended to group measure values into categories such as High, Medium, and Low. Confirm the exact calculation logic in Tableau before documenting thresholds.

Dashboard file

Packaged Tableau workbook: [Confirm the final .twbx filename and include it in the repository if size and licensing permit].

Dashboard URL: [Paste the verified Tableau Public dashboard URL here].

6. Report — Tableau Story

6.1 Story Design File

The Tableau Story is intended to guide the viewer through the analysis in a logical sequence. The story can begin with an overview, then move to adoption trends, country comparisons, usage by education level, impact by AI tool, and a comparison between students and educators. This sequence provides context before moving into more focused comparisons.

Story file and URL

Packaged workbook: [Confirm the final .twbx filename].

Tableau Story URL: [Paste the verified Tableau Public Story URL here].

7. Performance Testing

7.1 Utilization of Data Filters

Test each available filter independently and in combination. Confirm that selecting a year, country, region, or user type updates all intended charts and that clearing the filter restores the full view. Check that titles, legends, and labels remain readable after filtering.

7.2 Number of Calculation Fields

The project description mentions Adoption Category and Impact Category. Verify the actual workbook and record the final total below.

Verified calculation fields: [Enter the number and list all calculation fields].

7.3 Number of Visualizations

The visualizations described in this report include eight analysis views. Confirm the final count in the Tableau workbook before submission, as the actual workbook may differ.

Verified visualization count: [Enter the exact count from Tableau].

Additional tests

- Confirm the Flask home page opens without errors.
- Test the Dashboard and Story buttons and confirm they open the correct content.
- Check the Tableau links in a private/incognito browser window to verify public access.
- Check that the GitHub repository contains the intended files and does not contain passwords, API keys, or other secrets.
- Test the demo video URL in a signed-out browser window if the faculty needs to access it.

8. Conclusion / Observation

This project demonstrates an end-to-end data analytics workflow: working with a dataset, reviewing and managing data, developing visualizations in Tableau, organizing selected views into a Story, integrating the Tableau content into a Flask web interface, and maintaining project artifacts in GitHub. The dashboard is designed to support exploration of AI adoption, usage, infrastructure, impact, and barriers across the categories represented in the data.

The conclusions drawn from the charts depend on the quality and nature of the underlying dataset. If the dataset is synthetic or sample data, observations should be presented as patterns within that dataset, not as verified claims about global education systems.

9. Future Scope

- Replace or supplement sample data with validated datasets from reliable, cited sources.
- Automate data refresh and preprocessing where suitable data sources are available.
- Improve the Flask interface with responsive design, accessibility improvements, and clearer error messages.
- Add more detailed comparisons by country, education level, year, or AI tool if the data supports them.
- Add a data dictionary and documented definitions for every metric and calculated field.
- Add automated tests for the Flask routes and a repeatable checklist for Tableau filter testing.
- Publish a narrated demo video and include its verified URL in the project documentation.

10. Appendix

10.1 Source Code (if any)

The source code for the Flask website and supporting front-end files should be included in the GitHub repository. Common files may include app.py, templates/index.html, static CSS or JavaScript files, and requirements.txt. Only include filenames that actually exist in the final project.

10.2 GitHub & Project Demo Link

Resource	URL / status
GitHub repository	https://github.com/PrasadMhasal10/Global_AI_Adoption_In_Education
Tableau Dashboard	[Paste verified public dashboard URL]
Tableau Story	[Paste verified public story URL]
Narrated demo video	[Paste the video URL after uploading it]
Flask live website	[Add a deployed URL if available; localhost URLs work only on your own computer]

Submission checklist

- Replace all bracketed placeholders with verified details.
- Add your name, student ID, institution, and faculty name.
- Confirm the number of Tableau visualizations and calculation fields.
- Insert screenshots of the dataset, dashboard, story, and Flask interface if required by your faculty.
- Verify that the GitHub repository and demo video links open correctly.
- Review the final PDF before uploading it to GitHub.